

Five-Term Arithmetic Progressions and Cubes in a Pure Cubic Field

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Abstract

For a nonconstant rational five-term arithmetic progression, we ask how many nonzero terms can, after multiplication by one common rational scalar, be cubes in a single nontrivial pure cubic field. We prove that the exact maximum is four. The proof begins by determining the kernel of the map from rational cube classes to the Kummer group of $\mathbf{Q}(\sqrt[3]{D})$. After normalizing radicands and quotienting by affine color change and reversal, we obtain exactly 25 five-position color orbits. Ten orbits are excluded by known theorems on rational cubes in arithmetic progression. A prime-support lemma reduces the remaining 15 orbits to 60 explicit systems of three diagonal cubic equations. For every system, exhaustive finite-field enumeration gives an obstruction at a displayed prime of good reduction; a versioned certificate records all $p^2 - 1$ nonzero parameter pairs. The progression $(-3, -1, 1, 3, 5)$ supplies four counted terms over $\mathbf{Q}(\sqrt[3]{3})$. Thus $R_{(3,1)}^\times(5) = 4$. This finite classification excludes every five-hit color pattern but does not classify all rational progressions attaining four hits.

Keywords: arithmetic progression; pure cubic field; Kummer class; perfect cube; local obstruction.

Mathematics Subject Classification (2020): Primary 11B25; Secondary 11D41, 11R16, 11Y50.

1 Definition and normalizations

Let $A_i = a + id$ for $0 \leq i < 5$, where $a, d \in \mathbf{Q}$ and $d \neq 0$. For a nontrivial class $[D] \in \mathbf{Q}^*/\mathbf{Q}^{*3}$ put $K_D = \mathbf{Q}(\sqrt[3]{D})$. Define $R_{(3,1)}^\times(5)$ as the maximum number of nonzero A_i for which there are one D and one $\lambda \in \mathbf{Q}^*$ such that $\lambda A_i \in K_D^3$ at every counted position.

The radicand may be chosen as a positive cube-free integer: -1 is already a rational cube, and multiplying D by a rational cube changes no field. Moreover D and D^2 generate the same field. We quotient progressions by common rational scaling and by reversal $(a, d) \mapsto (a + 4d, -d)$. Zero has no class in $\mathbf{Q}^*/\mathbf{Q}^{*3}$ and is never counted. If D is a rational cube, then $K_D = \mathbf{Q}$ and the problem is the ordinary rational-cube problem, whose five-term maximum is three [2]; this degree-one degeneration cannot attain our maximum.

2 The pure-cubic Kummer kernel

Lemma 2.1. *If $[D] \neq 1$ in $\mathbf{Q}^*/\mathbf{Q}^{*3}$, then*

$$\ker(\mathbf{Q}^*/\mathbf{Q}^{*3} \longrightarrow K_D^*/K_D^{*3}) = \langle [D] \rangle = \{1, [D], [D]^2\}.$$

Equivalently, a nonzero rational r is a cube in K_D if and only if $[r] \in \langle [D] \rangle$.

Proof. Write $\alpha^3 = D$ and $y = a + b\alpha + c\alpha^2$. Reduction of y^3 in the basis $1, \alpha, \alpha^2$ gives nonconstant coefficients

$$3(a^2b + Dac^2 + Db^2c), \quad 3(a^2c + ab^2 + Dbc^2).$$

If $a = 0$, their vanishing implies $bc = 0$, so y lies on either the $\mathbf{Q}\alpha$ or $\mathbf{Q}\alpha^2$ axis. If $a \neq 0$, put $B = b/a$, $C = c/a$. The two equations are

$$B + DC^2 + DB^2C = 0, \quad C + B^2 + DBC^2 = 0.$$

Their resultants, eliminating C and B respectively, are

$$BD(B^3D - 1)^2, \quad C(C^3D^2 - 1)^2.$$

Since D is not a rational cube, the first forces $B = 0$, and then the first equation forces $C = 0$. Hence an element of K_D with rational cube lies in exactly one of $\mathbf{Q}$, $\mathbf{Q}\alpha$, $\mathbf{Q}\alpha^2$. Cubing proves the asserted three classes, and the converse is immediate. $\square$

Thus, after the common scaling, every counted term has a unique color $c_i \in \mathbf{F}_3$ and can be written

$$A_i = D^{c_i} x_i^3, \quad x_i \in \mathbf{Q}^*. \quad (1)$$

Changing the common scale by D^k , replacing D by D^2 , and reversing the progression act on color words by

$$c_i \mapsto c_i + k, \quad c_i \mapsto -c_i, \quad (c_0, \dots, c_4) \mapsto (c_4, \dots, c_0).$$

3 The 25 color orbits

Exact enumeration gives 25 orbits of $\mathbf{F}_3^5$ under these operations. For completeness, Burnside's lemma gives the same count without trusting the choice of representatives: the fixed-point counts of the 12 group elements are

$$243, 27, 9, 9, 9, 1, 1, 1, 0, 0, 0, 0.$$

Their sum is 300, and hence there are $300/12 = 25$ orbits. Nine contain a monochromatic three-term arithmetic progression of indices among

$$(0, 1, 2), (1, 2, 3), (2, 3, 4), (0, 2, 4).$$

Such an orbit would give three nonconstant rational cubes in arithmetic progression after scaling, contrary to Darmon–Merel [1]. Exactly one further orbit, represented by 00100, survives this test but has four terms of one color, at positions 0, 1, 3, 4. It contradicts the exact ordinary bound $P_5(3) = 3$ of Hajdu–Tengely [2]. The remaining representatives are listed below. There is no hidden integrality assumption here: choosing an integer L that clears the finitely many rational cube roots and the progression parameters, multiplication of the entire progression by the common cube L^3 makes both the progression and those roots integral. Cube positions and nonconstancy are therefore preserved.

$$\begin{aligned} &00101, 00102, 00110, 00112, 00120, 00121, 00122, 01001, 01002, 01012, \\ &01021, 01102, 01120, 01201, 01210. \end{aligned} \quad (1)$$

4 Prime support and the 60 curves

Clear denominators and divide by the common gcd, obtaining a primitive integer progression $A_i = a + id$. Its classes have the affine form $g + c_i\delta$ in $\mathbf{Q}^*/\mathbf{Q}^{*3}$. For $i \neq j$,

$$\gcd(A_i, A_j) \mid |i - j|,$$

because $\gcd(a, d) = 1$. Therefore a prime $p > 3$ divides at most one of the five terms.

Lemma 4.1 (prime-support lemma). *For each of the 15 words in (1), every $p > 3$ occurs in every A_i with exponent divisible by three. Consequently both g and δ are supported only at 2 and 3.*

Proof. If a $p > 3$ exponent were nonzero modulo three, the five valuation residues would be supported at one position. But they must be the values of the affine function $c \mapsto g_p + c\delta_p$ on the displayed color word. For a two-color word in (1), both color blocks have size at least two, so a singleton support is impossible. For a three-color word, an affine function over $\mathbf{F}_3$ vanishing at the other two colors vanishes identically. This is again impossible. Hence all five residues vanish. Taking differences of two used colors proves the assertion for δ , and then for g . $\square$

After another common rational scaling removes g , the four possible one-dimensional subspaces of the $\{2, 3\}$ -supported class plane have representatives

$$D = 2, 3, 6, 18.$$

For a word $c = (c_0, \dots, c_4)$ and one such D , all five hits give the concrete projective curve in $\mathbf{P}^4$

$$D^{c_i} x_i^3 - 2D^{c_{i+1}} x_{i+1}^3 + D^{c_{i+2}} x_{i+2}^3 = 0 \quad (0 \leq i \leq 2). \quad (2)$$

At every prime used below, $p \geq 7$ and $p \nmid D$. These conditions really give good reduction. Put $u_i = D^{c_i} x_i^2$. After deleting the invertible factor 3, the Jacobian matrix of the three equations is

$$\begin{pmatrix} u_0 & -2u_1 & u_2 & 0 & 0 \\ 0 & u_1 & -2u_2 & u_3 & 0 \\ 0 & 0 & u_2 & -2u_3 & u_4 \end{pmatrix}.$$

A projective solution has at most one $u_i = 0$: two zero terms in the induced five-term progression would force both its initial term and common difference to vanish because $p > 4$. If no u_i is zero, columns 0, 1, 2 give a nonzero minor. If the unique zero is respectively u_0, u_1, u_2, u_3, u_4 , columns 124, 024, 014, 014, 013 give

$$3u_1u_2u_4, -2u_0u_2u_4, u_0u_1u_4, u_0u_1u_4, -2u_0u_1u_3.$$

Thus the Jacobian has rank 3 at every geometric point, proving smoothness.

Table 1 gives, in word order (1), a prime with no nonzero projective point on (2); columns correspond to $D = 2, 3, 6, 18$. Every listed prime is coprime to $3D$. The certificate enumerates all $(a, d) \in \mathbf{F}_p^2 \setminus \{(0, 0)\}$ and

word	2	3	6	18	word	2	3	6	18
00101	7	7	31	7	00102	7	7	13	7
00110	7	7	13	7	00112	7	7	13	7
00120	7	13	13	31	00121	7	19	61	31
00122	13	13	13	43	01001	7	7	13	7
01002	7	7	61	7	01012	13	7	13	7
01021	7	13	13	31	01102	13	7	13	7
01120	7	7	61	7	01201	7	7	13	7
01210	7	13	31	31					

Table 1: Good primes obstructing the 60 remaining word–radicand models

tests whether $a + id \in D^{c_i} \mathbf{F}_p^3$ for all five positions. This is equivalent to enumerating the projective solutions of (2), including solutions with some zero coordinates. Absence modulo p excludes a rational point.

The complete finite computation is identified by

```
PAPER_CUBE_KUMMER5_CERTIFICATE.json,
schema paper-cube-pure-cubic-kummer-n5-v2,
SHA-256
7c5ff0bd36ebfc3bace0ca5898625b2fd7f03d318c754c6650d3a2483dd83977.
```

Its generator and tests are located by the root-level file

```
PAPER_CUBE_KUMMER5_SUPPLEMENT_MANIFEST.json.
```

Each of the 60 cells scans all $p^2 - 1$ nonzero parameter pairs, for 23520 pairs in total. The local supplement is not a public archive; at submission its manifest must be deposited at the immutable DOI or URL stated in the data-availability record.

5 Exact maximum and scope of classification

Theorem 5.1. *With nonzero terms and nontrivial pure cubic fields as in Section 1,*

$$R_{(3,1)}^\times(5) = 4.$$

Proof. A five-hit progression has one of the 25 color orbits. The first ten are excluded in Section 3. Lemma 4.1 and the 60 good-prime obstructions exclude the remaining 15. Hence the maximum is at most four. For the reverse inequality, in $K = \mathbf{Q}(\alpha)$ with $\alpha^3 = 3$, the nonzero progression

$$-3, -1, 1, 3, 5$$

has its first four terms equal to the cubes of $-\alpha, -1, 1, \alpha$. □

The theorem includes a complete classification and exclusion of every *five-hit color/position class*. It does not classify every rational progression attaining four hits. That latter set is a distinct Diophantine problem: already a four-position pattern is naturally a weighted complete intersection, and several patterns have multiple rational examples. We do not replace that rational-point classification by a bounded search.

If zero were counted, $-2, -1, 0, 1, 2$ would give five hits in $\mathbf{Q}(\sqrt[3]{2})$; this is exactly why the superscript $\times$ is necessary.

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Data and code availability. The exact source, tests, and finite certificate will be deposited with the accepted release at PERSISTENT ARCHIVE DOI/URL -- HUMAN AUTHOR INPUT REQUIRED. The local release candidate is not yet a public archive.

Use of generative AI. Generative AI tools were used during exploratory mathematical research, drafting, code generation, and preparation of the reproducibility package. No AI system is listed as an author. Before submission, the human authors must independently verify every statement, proof, computation, citation, and file, approve that this disclosure is complete, and accept full responsibility for the submitted work.

Ethics approval, consent to participate, and consent for publication. Not applicable: this is a mathematical study involving no human participants, animals, or personal data.

Supplementary information. Online Resource 1, PAPER_CUBE_KUMMER5_ARCHIVE_CANDIDATE.zip (ZIP), contains the exact source, PDF, generator, tests, 60-model certificate, manifest, data dictionary, and release metadata used for the finite classification. It is a local candidate until the human authors approve and deposit the immutable archive.

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